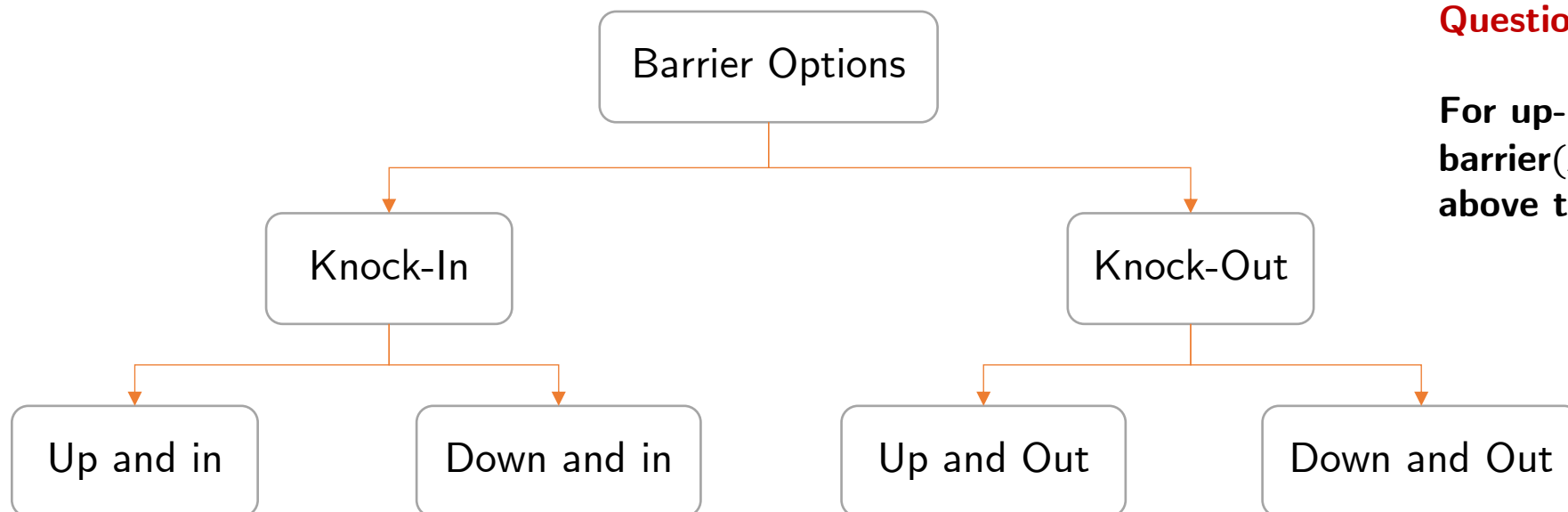


Barrier Options

(1. First Passage Times)

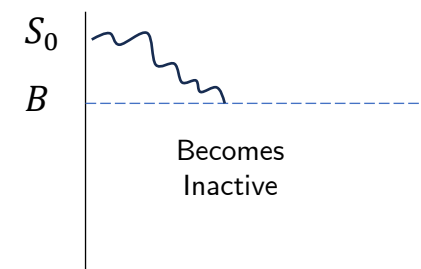
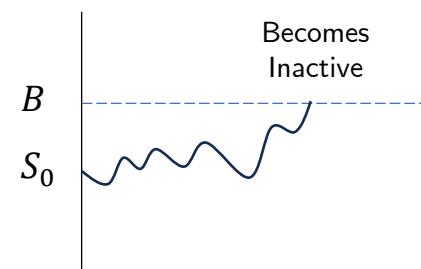
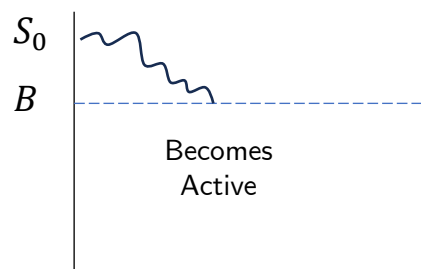
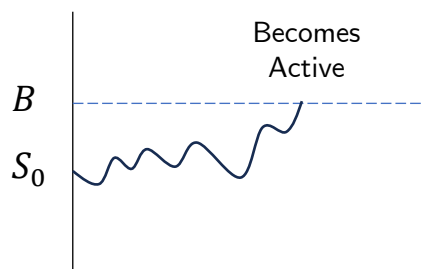
Barrier Options

Barrier options come in the following flavors



Question

For up-and-out Call, the barrier(B) should be kept above the strike (K). (T/F) ?



MGF and PGF

We spoke about **MGF (Moment Generating Function)** earlier. There is a related power series applicable for discrete random variables, called as PGF (Probability Generating Functions)

$$\textbf{(MGF)} \quad M_X(t) = \mathbb{E}[e^{tX}] = 1 + t\mathbb{E}[X] + \frac{1}{2}t^2\mathbb{E}[X^2] + \dots$$

$$\mathbb{E}[X] = \left. \frac{dM}{dt} \right|_{t=0} \quad \mathbb{E}[X^2] = \left. \frac{d^2M}{dt^2} \right|_{t=0}$$

$$\begin{aligned} \textbf{(PGF)} \quad G_X(z) = \mathbb{E}[z^X] &= \sum_{x=0}^{\infty} p(x)z^x = z^0P(X=0) + z^1P(X=1) + z^2P(X=2) + \dots \\ &= P(X=0) + zP(X=1) + z^2P(X=2) + \dots \end{aligned}$$

The coefficient of z^n , denotes the probability of X taking a value n , i.e., $P(X=n)$

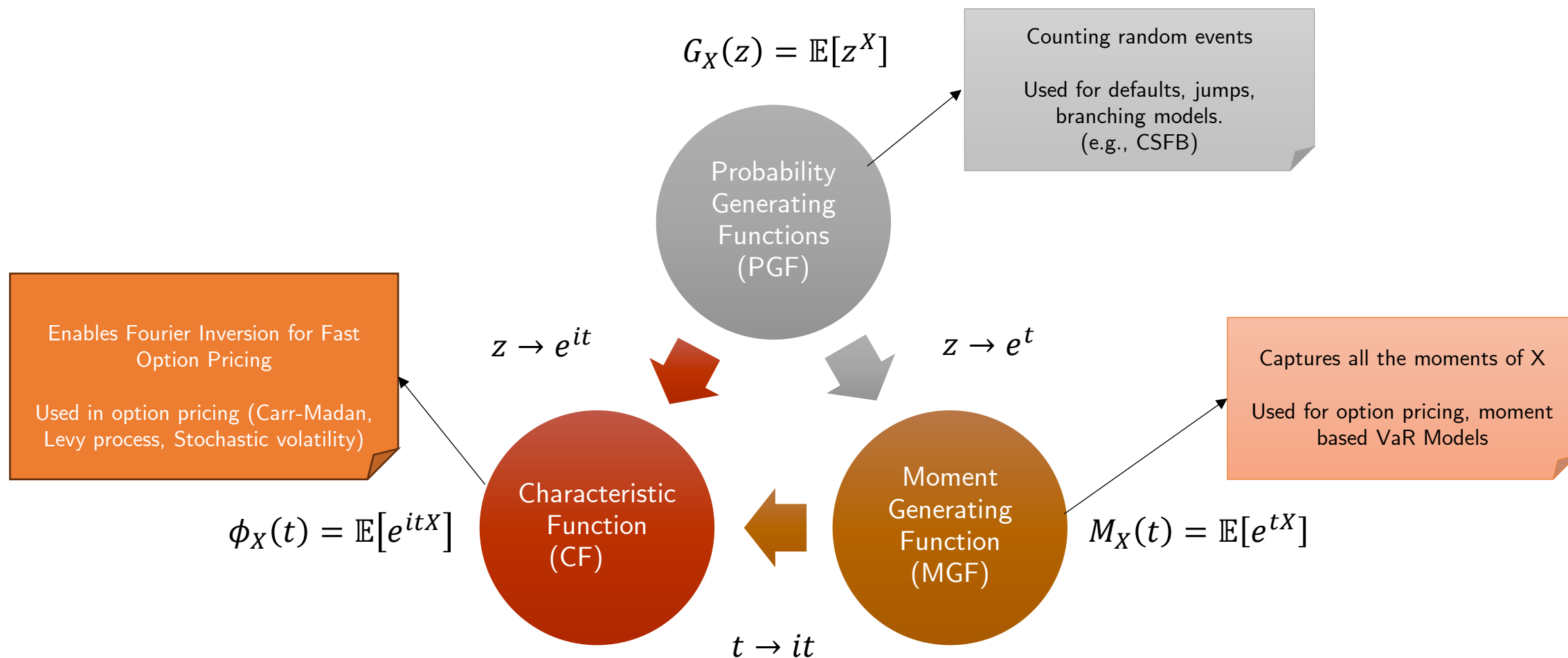
Properties of PGF

Here are some important properties of PGF

$$G_X(z) = \mathbb{E}[z^X] = P(X = 0) + zP(X = 1) + z^2P(X = 2) + \dots$$

- 1) The probability mass function of X is recovered by taking derivative of G , $P(X = k) = \frac{G^{(k)}(0)}{k!}$
- 2) Total probability = 1, $G_X(1) = 1$
- 3) Expectation can be recovered as $\mathbb{E}[X] = G'(1)$
- 4) Sum of **independent randoms** $G_{X_1+X_2}(z) = G_{X_1}(z).G_{X_2}(z)$
- 5) PGF uniquely determines the probability distribution of X

MGF, PGF, CF



Exercise



- 1) Find the PGF of
 - a) Bernoulli Random and
 - b) Binomial Random

- 2) Find the PGF of Poisson distribution. Derive the Expectation from PGF.

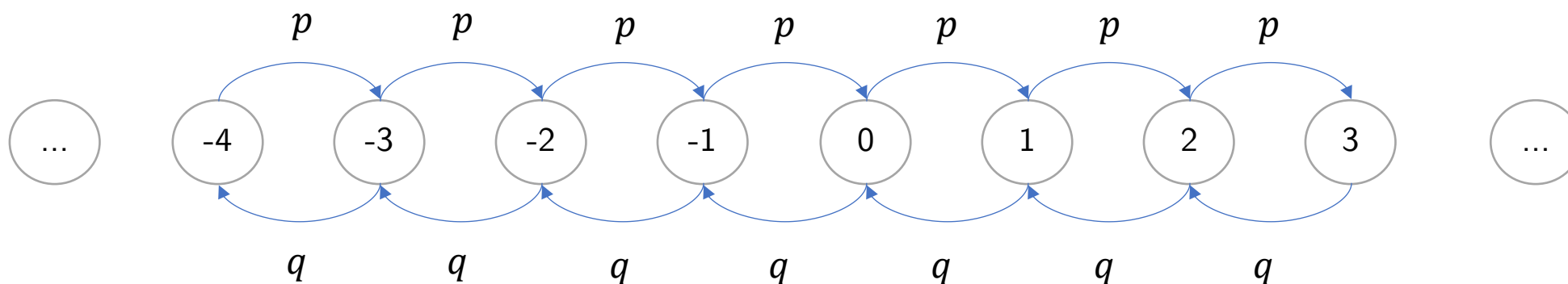
Random Walk



Recall that a random walk is a series of i.i.d steps where the increments can be either $+1$ or -1 .

(i.i.d increments)
$$X_i = \begin{cases} 1, & \text{prob} = p \\ -1, & \text{prob} = q \end{cases} \quad p + q = 1$$

Walker
$$S_n = X_1 + X_2 + \cdots + X_n, \quad S_0 = 0$$



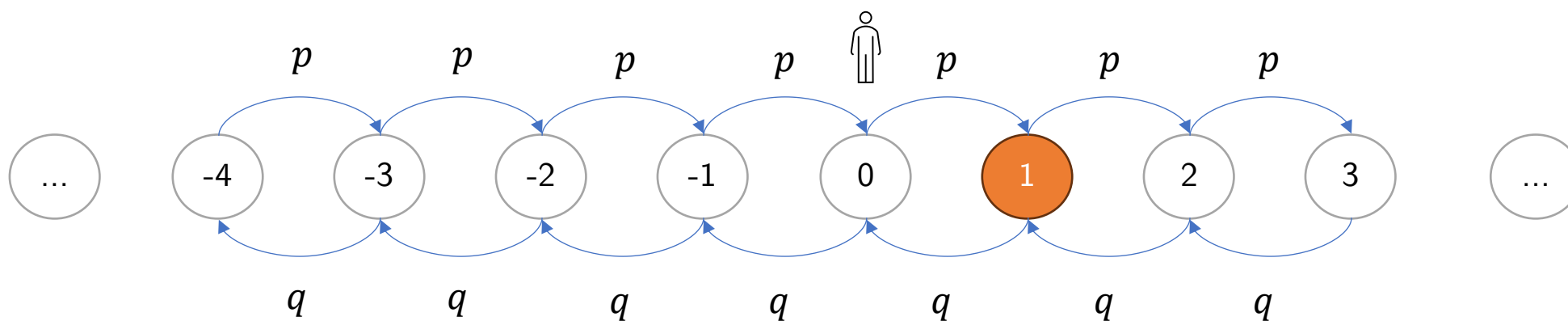
Show that the PGF of the walk is $G_{S_n}(z) = (zp + qz^{-1})^n$

First Passage Time (I)

Given the Simple Random Walk, we want to find out the following

The distribution of first time the walker reaches +1. This is called as 'hitting time' or 'first passage time'

Define the stopping time as $\tau = \min \{S_k = 1 | S_0 = 0\}$. Let's derive the PGF of τ



First Passage Time (II)

The PGF of τ is given as $G_\tau(z) = \mathbb{E}[z^\tau]$

$$\tau = \min \{S_k = 1 | S_0 = 0\}$$

Let's do one-step conditioning $G_\tau(z) = \mathbb{E}[\mathbb{E}(z^\tau | \text{one step})]$

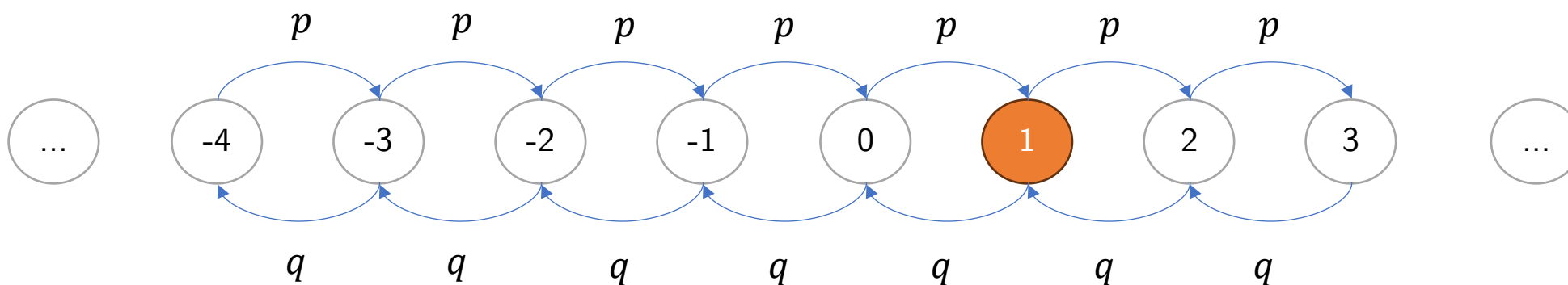
$$= p[\mathbb{E}(z^\tau | \text{right})] + q[\mathbb{E}(z^\tau | \text{left})]$$

$$= pz^1 + q[\mathbb{E}[z^{1+\tau'+\tau''}]]$$

τ' and τ'' are the copies of τ indicating
Hitting times from -1 to 0 and 0 to 1

$$= pz + qzG_\tau(z)^2$$

Characteristic equation : $\boxed{qz\alpha^2 - \alpha + pz = 0}$ where $\alpha = G_\tau(z)$



First Passage Time (III)

Characteristic equation : $qz\alpha^2 - \alpha + pz = 0$

Solving $\alpha = \frac{1 \pm \sqrt{1 - 4pqz^2}}{2qz}$

Since α is a PGF, we need to ensure it does not blow up for small values of z . We must take

$$G_\tau(z) = \frac{1 - \sqrt{1 - 4pqz^2}}{2qz}$$

For simple symmetric random walk, where $p = q = \frac{1}{2}$

$$G_\tau(z) = \frac{1 - \sqrt{1 - z^2}}{z}$$

Probability of hitting

Are we guaranteed to hit +1 in finite time ? In other words, does the first passage occurs at all ? We are looking for probability $P(\tau < \infty)$. Note that $P(\tau < \infty) = G_\tau(1)$. Using $p + q = 1$

$$\boxed{G_\tau(1) = \frac{1 - \sqrt{1 - 4pq}}{2q}} \rightarrow \boxed{G_\tau(z) = \frac{1 - |p - q|}{2q}}$$

$p < q$ $p > q$

$\downarrow$
 $G_\tau(1) = \frac{p}{q}$

$\downarrow$
 $G_\tau(1) = 1$

If $p > q$ (in other words, there is a +ve drift in the process), then the hitting +1 is guaranteed. However, if $p < q$ (negative drift), then there is non-zero probability $\left(1 - \frac{p}{q}\right)$ that $\tau = \infty$ (first passage never occurs)



General First Passage Times

What is the distribution of hitting times for a level $= +m$?

Define $\tau_m = \min \{S_k = m | S_0 = 0\}$

The PGF is given as

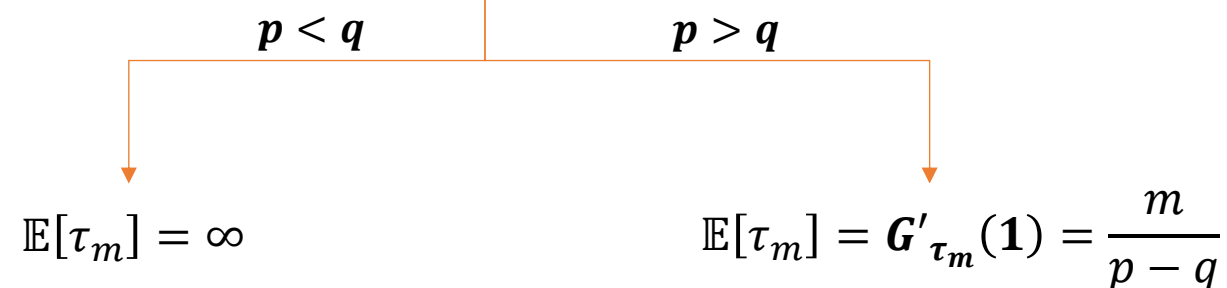
$$G_{\tau_m}(z) = \left(\frac{1 - \sqrt{1 - 4pqz^2}}{2qz} \right)^m$$

Expectation : $\mathbb{E}(\tau_m)$

What is the expected value of the hitting time for the barrier = m ? Since, we have the PGF, we can get the expected hitting time by taking

$$\mathbb{E}[\tau_m] = G'_{\tau_m}(1)$$

$$G_{\tau_m}(z) = \left(\frac{1 - \sqrt{1 - 4pqz^2}}{2qz} \right)^m$$



The diagram shows a central box containing the PGF $G_{\tau_m}(z) = \left(\frac{1 - \sqrt{1 - 4pqz^2}}{2qz} \right)^m$. Two orange arrows branch out from the bottom of this box. The left arrow is labeled $p < q$ and points to the equation $\mathbb{E}[\tau_m] = \infty$. The right arrow is labeled $p > q$ and points to the equation $\mathbb{E}[\tau_m] = G'_{\tau_m}(1) = \frac{m}{p - q}$.

$$\mathbb{E}[\tau_m] = \infty \quad \text{for } p < q$$
$$\mathbb{E}[\tau_m] = G'_{\tau_m}(1) = \frac{m}{p - q} \quad \text{for } p > q$$

Probability : $\mathbb{P}(S_n = m)$

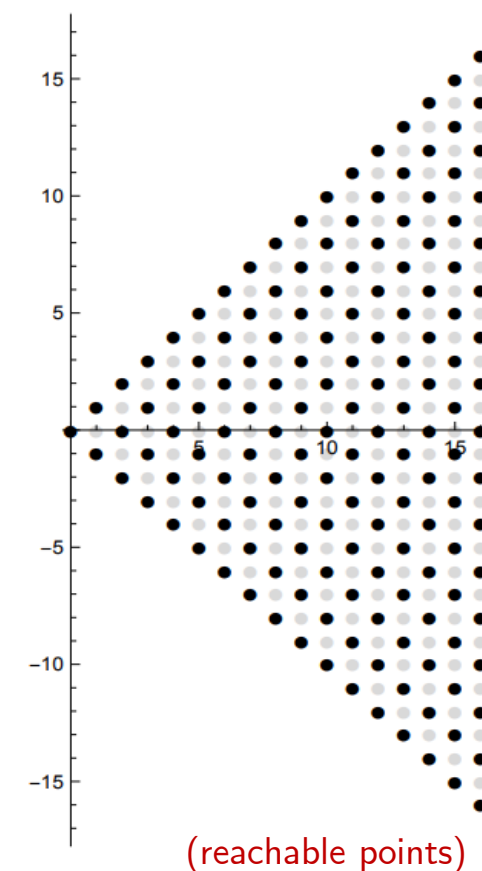
What is the probability the walker will be at level (m) after (n) steps ? Let's answer a simpler question first. Is it possible to reach level m after n steps ? If yes, what is the probability of reaching it ?

Define $N_{n,m}$ = (Total paths that reach level m after n steps). Then

$$N_{n,m} = \binom{n}{\frac{n+m}{2}}$$

$$\mathbb{P}(S_n = m) = P_{n,m} = \binom{n}{\frac{n+m}{2}} p^{\frac{n+m}{2}} q^{\frac{n-m}{2}}$$

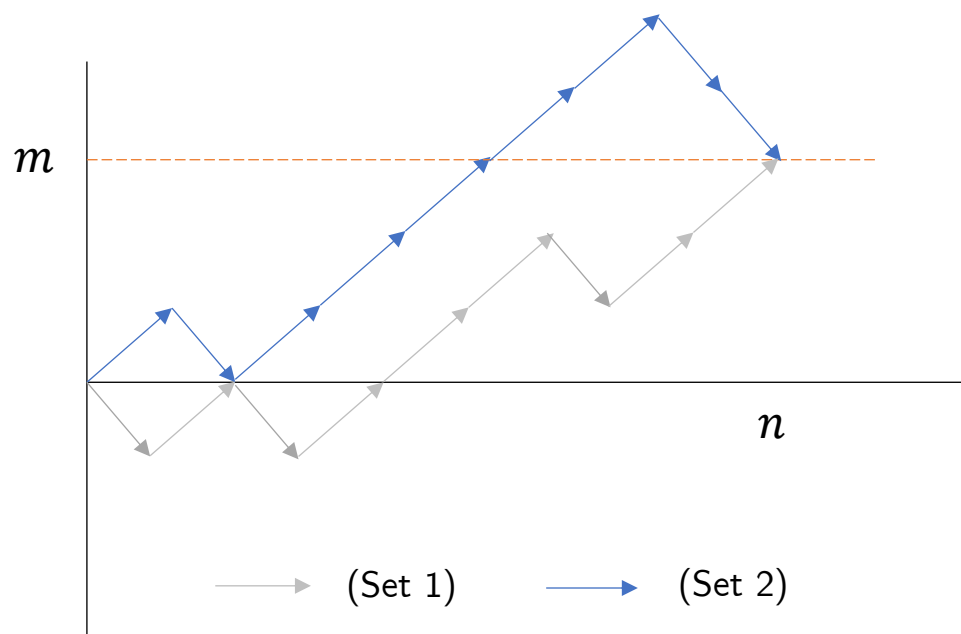
Note that the reachability condition is given as $n \equiv m \pmod{2}$



Probability : $\mathbb{P}(\tau_m = n)$

All the paths that lead from origin to level = m , in n steps i.e., $N_{n,m}$ consist of two sets of paths

- 1) **(Set 1)** Paths which visit level = m for the first time at step = n (This is our hitting time)
- 2) **(Set 2)** Paths which visits level = m before step = n and then return to level = m at step = n



The probability of first passage (level m) taking a value n is a fraction of the probability of being at level m at time n . The formula is given by **Ballot principle**.

$$\mathbb{P}(\tau_m = n) = \left(\frac{m}{n}\right) \mathbb{P}(S_n = m)$$

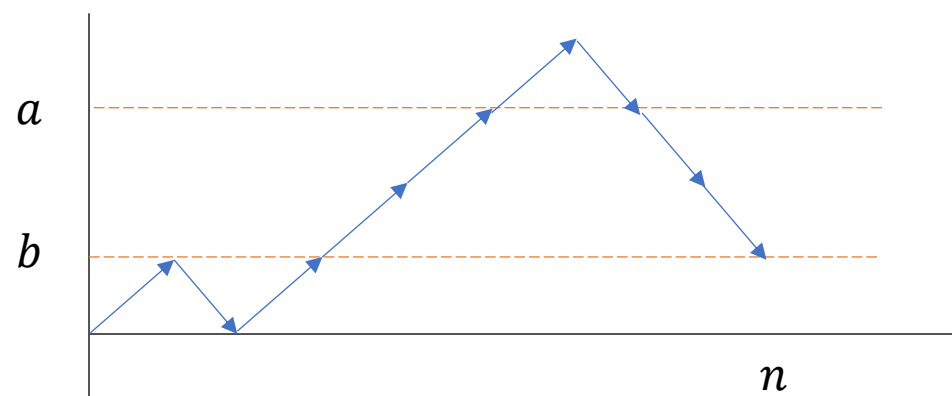


Reflection Principle (I)

Consider a simple symmetric random walk. What is the probability that the walker crosses the barrier a at some interim step but ends at $b \leq a$ after n steps ? In other words, define a running maximum

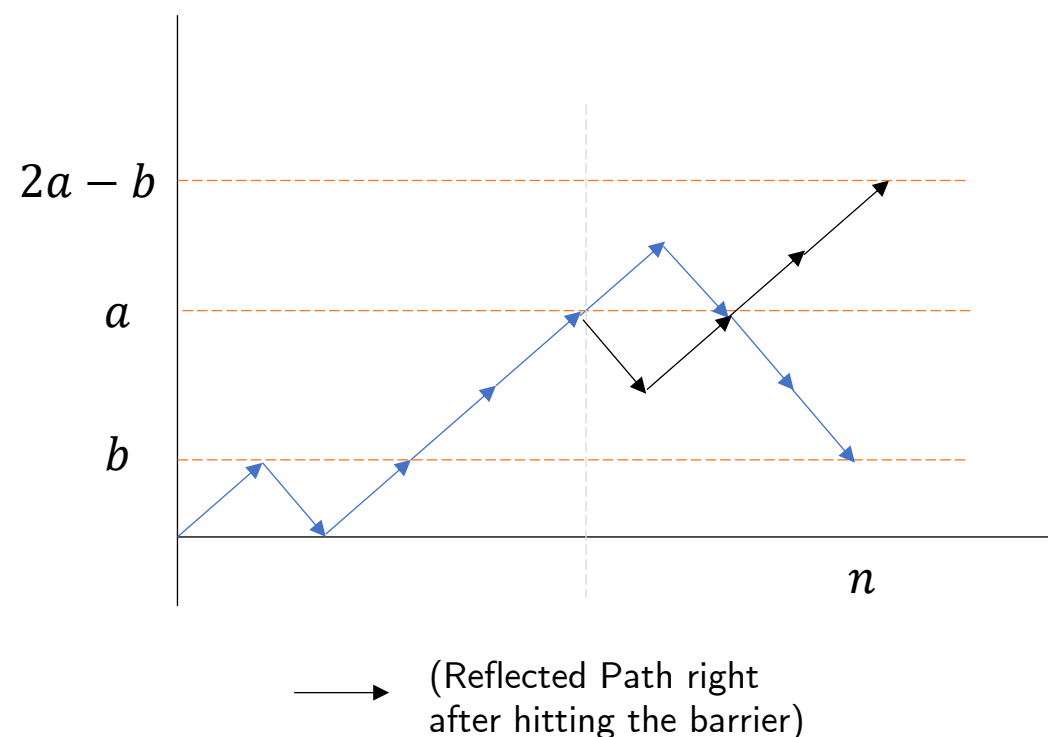
$$S_n^* = \max_{1 \leq j \leq n} S_j$$

Find the probability $\mathbb{P}(S_n^* \geq a, S_n = b)$



Reflection Principle (II)

Consider a simple symmetric random walk. What is the probability that the walker crosses the barrier a at some interim step but ends at $b \leq a$ after n steps ? In other words, define a running maximum



There is a bijection between the original paths that satisfy the above question and the reflected paths right after hitting the barrier a all of which must end at $2a - b$

$$\mathbb{P}(S_n^* \geq a, S_n = b) = \mathbb{P}(S_n = 2a - b)$$